Q.1 $p(x|\mu) = \mu^x (1-\mu)^{1-x}$; $x \in \{0,1\}$

$p(0|\mu) = 1-\mu$
$p(1|\mu) = \mu$

i) $E[x] = \sum_{x=\{0,1\}} p(x|\mu) \cdot x$

$= 0 \times p(0|\mu) + 1 \times p(1|\mu)$

$= 0 \times (1-\mu) + 1 \times \mu$

$= \mu$

ii) $E[x^2] = 0^2 \times (1-\mu) + 1^2 \times \mu = \mu$

$var[x] = E[x^2] - (E(x))^2 = \mu - \mu^2 = \mu(1-\mu)$

iii) $H(x) = -\sum_{x \in \{0,1\}} p(x|\mu) \log p(x|\mu)$

$= -\mu \log \mu - (1-\mu) \log (1-\mu)$

Q.2 kappa score $= \dfrac{\sum_c \left[p(y_1 = c, y_2 = c) - p(y_1 = c)\, p(y_2 = c) \right]}{1 - \sum_c \left[p(y_1 = c)\, p(y_2 = c) \right]}$

$= \dfrac{\left[\frac{4}{10} - \frac{8}{10} \times \frac{5}{10} \right] + \left[\frac{1}{10} - \frac{2}{10} \times \frac{5}{10} \right]}{1 - \left[\frac{8}{10} \times \frac{5}{10} + \frac{2}{10} \times \frac{5}{10} \right]}$

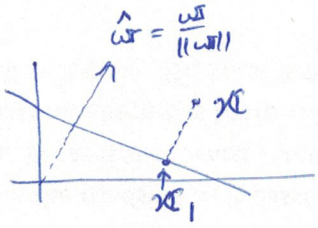
$= \dfrac{\frac{5}{10} - \frac{40+10}{100}}{1 - \frac{40+10}{100}} = 0$

Q.3 i) $\sum_{i=\{1,3,5,\dots\}} (x^T u_i)\, u_i$

ii) $(x^T x)^{1/2}$

iii) $\perp$ distance of x from line

$= \dfrac{w^T x + w_0}{\|w\|}$



$\therefore x_1 = x - \dfrac{w}{\|w\|} \times \left(\dfrac{w^T x + w_0}{\|w\|} \right)$

Q.6

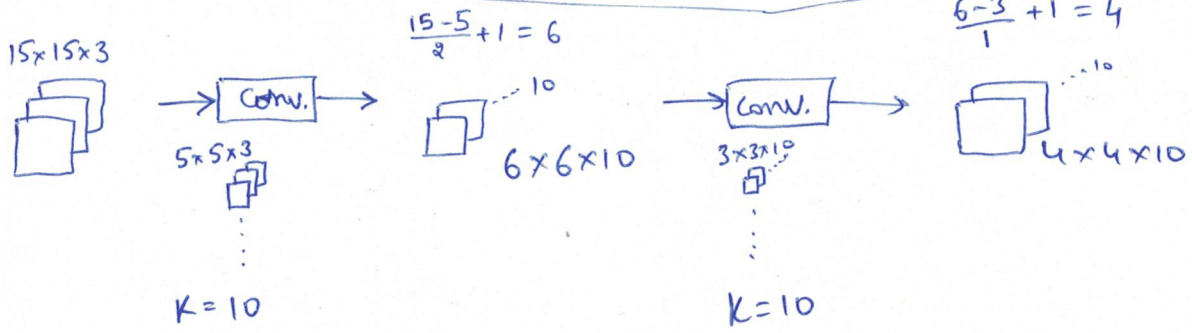
$$\text{output size} = \frac{N - F + 2P}{S} + 1$$

given: output size $= N$, $N = 15$, $F = 11$, $P = ?$, $S = 1$

$$\cancel{N} = \cancel{N} - F + 2P + 1$$

$$\therefore P = \frac{F-1}{2} = 5$$

Q.8



i) Total no. of learnable params $= (5 \times 5 \times 3 + 1) \times 10$ (bias)

$+ (3 \times 3 \times 10 + 1) \times 10$

$= 760 + 910 = 1670$

or 1650 (without bias)

iii) No. of multiplications $= (5 \times 5 \times 3) \times (6 \times 6 \times 10) + (3 \times 3 \times 10) \times (4 \times 4 \times 10)$

$= 41400$

Q.9

$$\begin{bmatrix} 0 & 1 & 0 \\ 4 & 0 & 3 \\ 0 & 9 & 7 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 2 & 0 & 5 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\xrightarrow{\text{Conv.}} \begin{bmatrix} 9 & 0 \\ 0 & 21 \end{bmatrix} + \begin{bmatrix} 13 & 9 \\ 12 & 23 \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ 12 & 44 \end{bmatrix} \xrightarrow{\text{avg. pooling}} \frac{87}{4} = 21.75$$

Q.10 i)

$\pi \rightarrow K-1$ ($\because p(z)$ sums to 1)

$\mu \rightarrow K \times N$

$\Sigma \rightarrow K \times N^2$

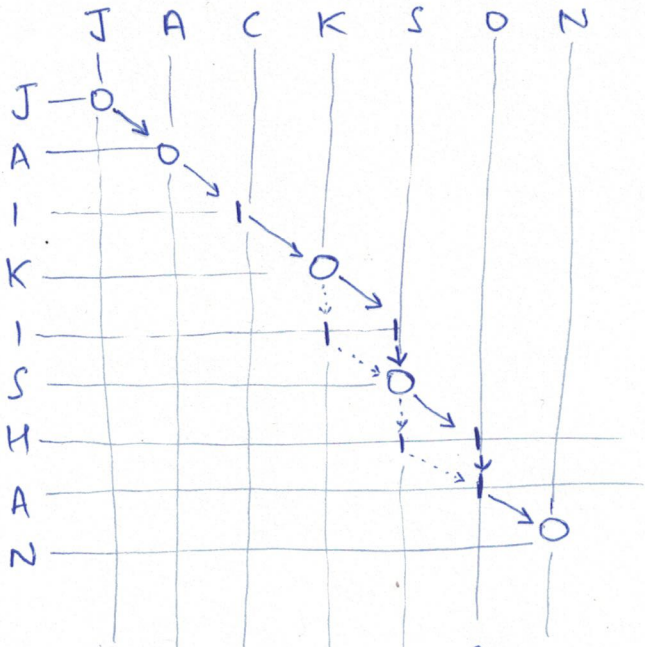
total $= K - 1 + KN + KN^2$

ii) $\Sigma \rightarrow K \times N$

(diagonal matrix)

total $= K - 1 + KN + KN$

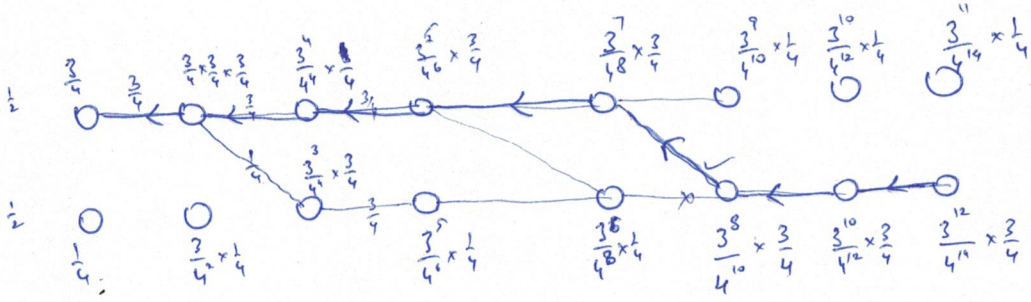
Q.11 ii)



min. cost = 4

best matched: J A I K S H N

Q.16



Q.22 p-value $= \left[{}^8C_0 \left(\frac{1}{2}\right)^8 + {}^8C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^7 + {}^8C_2 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^7 \right] \times 2$

(we have two − ves and six + ves $\Rightarrow b = 2$)

$$= \left(1 + 8 + \frac{\overset{4}{\cancel{8}} \times 7}{\cancel{2}}\right) \frac{1}{2^8} \times 2$$

$$= 37 \times \frac{1}{2^7}$$